

Date	27 JUNE 2026
Team ID	SWTID-2026-6540
Project Name	Opti-Crop: Smart Agricultural Production Optimization Engine
Maximum Marks	

Purpose of the Project

The purpose of **OptiCrop: Smart Agricultural Production Optimization Engine** is to develop an intelligent crop recommendation system that helps farmers select the most suitable crop based on soil nutrients and environmental conditions. The project leverages machine learning techniques to analyze agricultural data and provide accurate crop recommendations, enabling farmers to make informed decisions before cultivation.

OptiCrop aims to improve agricultural productivity by reducing the uncertainty involved in crop selection. By considering parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall, the system identifies the crop that is best suited to the given conditions. This helps increase crop yield while minimizing resource wastage.

The project also promotes sustainable farming by encouraging the efficient use of water, fertilizers, and other agricultural resources. It reduces the dependence on

traditional guesswork and manual decision-making, providing farmers with a simple, fast, and reliable AI-based solution.

Additionally, the project demonstrates the practical application of machine learning and web technologies in agriculture. It provides an easy-to-use web interface that makes advanced technology accessible to farmers and supports the adoption of data-driven farming practices.